

Decode
2026

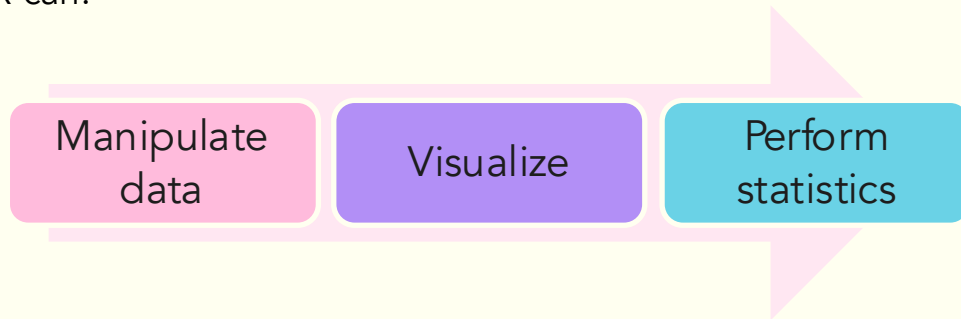
An Introduction to
R and Data
Wrangling with the
Tidyverse

What is R?

2

- R is a programming language and environment for statistical computing and graphics
- It was developed in 1991
- R is both **open source** and **open development**

R can:



Why R?

3

- Free and open source
- High level language
- Powerful and flexible
- Extensive add-on software (aka packages)
- Strong community!

What is RStudio?

4

- An Integrated Development Environment (IDE) for R
- Helps you:

Write code

View the
output of your
code

Find errors

Manage files

Interactive
tools

View
documentation



<https://posit.co/downloads/>

<https://posit.co/products/open-source/rstudio>

What is Posit Cloud?

5

- Platform that lets you use RStudio in a browser
- Removes the need to download software
- Lets you easily share projects for analysis and collaboration



<https://posit.cloud/>

Outline

6

Data Manipulation and Visualization

A tour
of R
studio

Understanding
R packages

Importing
data

Visualizing
Data

Merging
Data

Introduction
to
MOTRPAC
data

Creating
Objects

Manipulating
Data

Reshaping
Data

Let's get started!

7

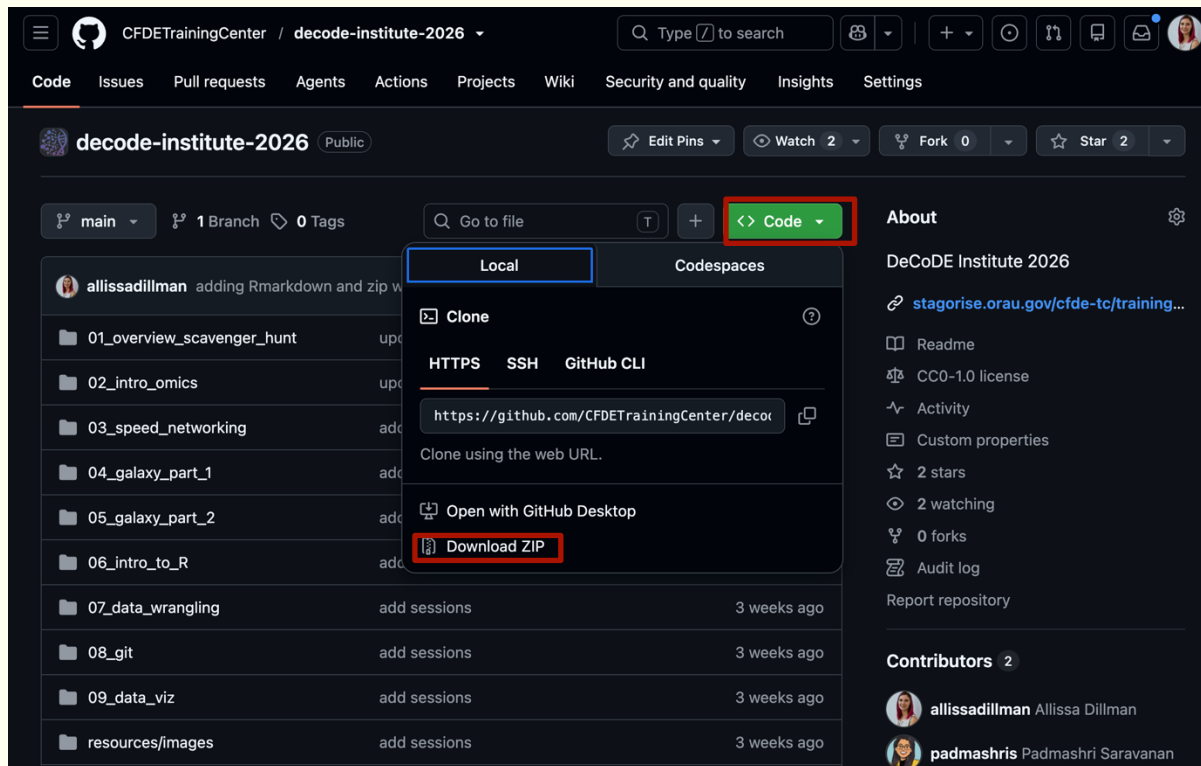
- Log in to: <https://cfdeworkspace.org/>
- If you do not have an account this may take time a backup is <https://usegalaxy.org/>
- Click interactive Tools
- Click RStudio
- Click Run Tool
- Click View on the side panel to open RStudio

The screenshot displays the Galaxy CFDE Cloud web interface. On the left sidebar, the 'Interactive Tools' icon is highlighted with a red box. The main panel shows the 'Interactive Tools' section with a search bar and a list of tools. The 'RStudio R 4.4.2 with Bioconductor 3.29' tool is selected and highlighted with a red box. Below the tool list, the 'Run Tool' button is also highlighted with a red box. The right panel shows the 'Tool Parameters' for the selected tool, including options for 'Include data into the environment', 'Additional Options', 'Email notification', 'Attempt to re-use jobs with identical parameters?', and 'Output Tags'. The bottom right panel shows the 'History' view with a list of datasets, including one labeled '11: RStudio' which is highlighted with a red box.

Let's get started!

8

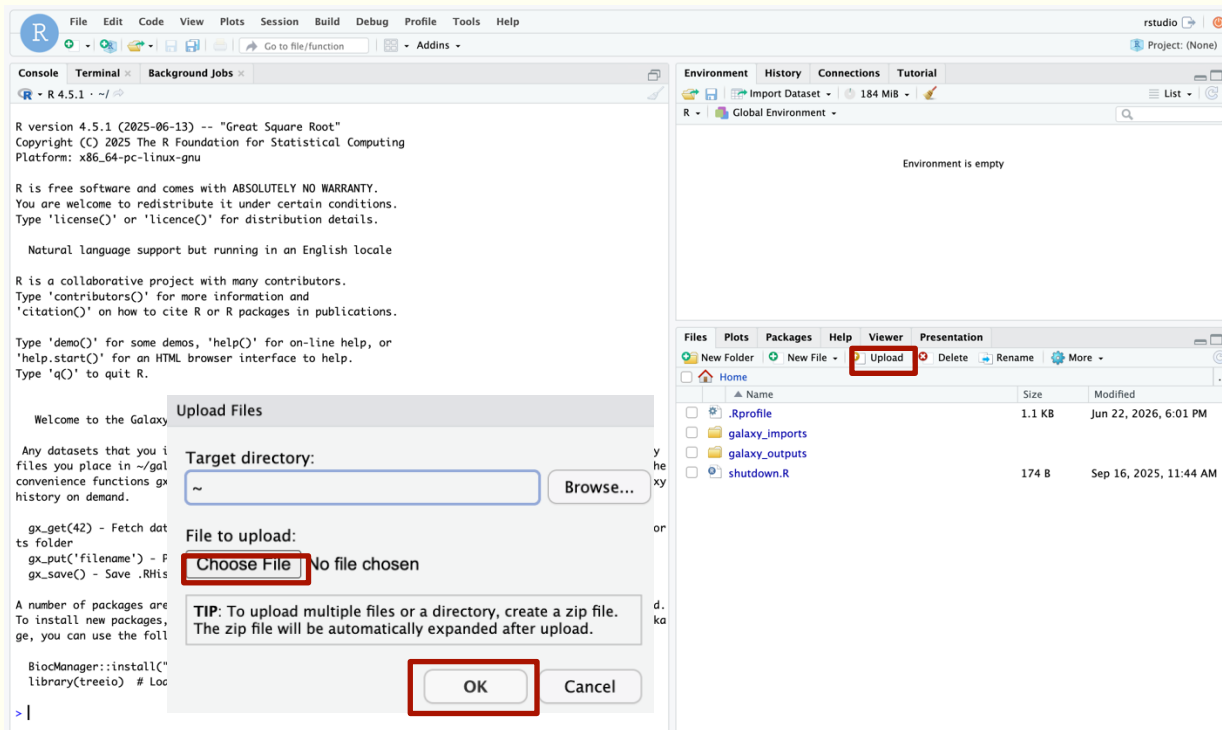
- <https://github.com/CFDETrainingCenter/decode-institute-2026>
- Click Code
- Click Download Zip



Let's get started!

9

- Click on Upload
- Choose the Rmd file we downloaded from github
- Click Ok
- Repeat for zip file



The screenshot displays the RStudio environment. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. The top right shows 'rstudio' and 'Project: (None)'. The left sidebar contains 'Console', 'Terminal', and 'Background Jobs'. The main console area shows the R version 4.5.1 (2025-06-13) and the 'Great Square Root' project. The right sidebar contains 'Environment', 'History', 'Connections', and 'Tutorial'. The 'Environment' panel shows 'Global Environment' and 'Environment is empty'. The bottom panel shows 'Files', 'Plots', 'Packages', 'Help', 'Viewer', and 'Presentation'. The 'Files' panel shows a file explorer view with a table of files.

Console Output:

```
R version 4.5.1 (2025-06-13) -- "Great Square Root"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Welcome to the Galaxy RStudio with Bioconductor Interactive Tool.

Any datasets that you included when starting RStudio are available in ~/galaxy_inputs directory. Any
files you place in ~/galaxy_outputs will be available in Galaxy once you stop RStudio. You can use th
e convenience functions gx_put(), gx_get(), and gx_save() to fetch and place data to your current Gal
axy history on demand.

gx_get(42) - Fetch dataset 42 from your Galaxy history. The file will be available in ~/galaxy_impo
rts folder
gx_put('filename') - Push a dataset to Galaxy
gx_save() - Save .RHistory, .RData to your Galaxy environment

A number of packages are pre-installed, which you can inspect with the 'installed.packages()' comman
d. To install new packages, you can use the BiocManager package. For example, to install the treeio p
ackage, you can use the following command to install it and load it:

BiocManager::install("treeio")
library(treeio) # Load the treeio package after installation

> |
```

Environment Panel:

Environment is empty

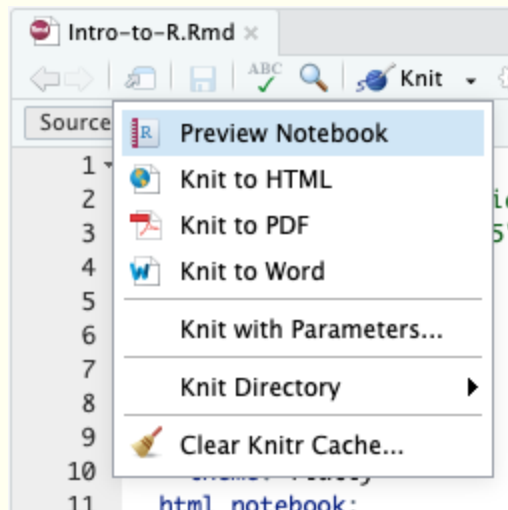
Files Panel:

Name	Size	Modified
.Rprofile	1.1 KB	Jun 27, 2026, 9:53 AM
galaxy_imports		
galaxy_outputs		
shutdown.R	174 B	Sep 16, 2025, 11:44 AM

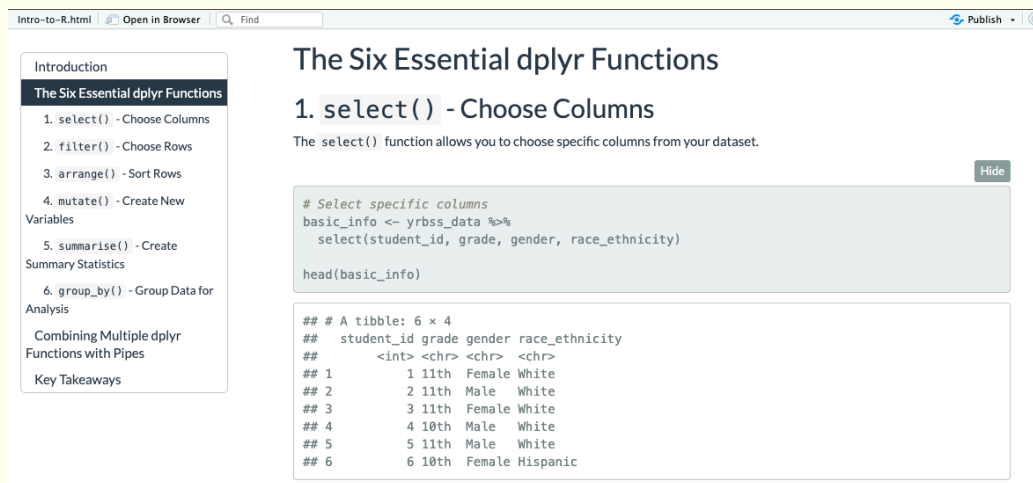
Interactive HTML files

12

Click on the dropdown and
select *Knit to HTML*



This will create a HTML file viewable in any browser without
needing any coding platform

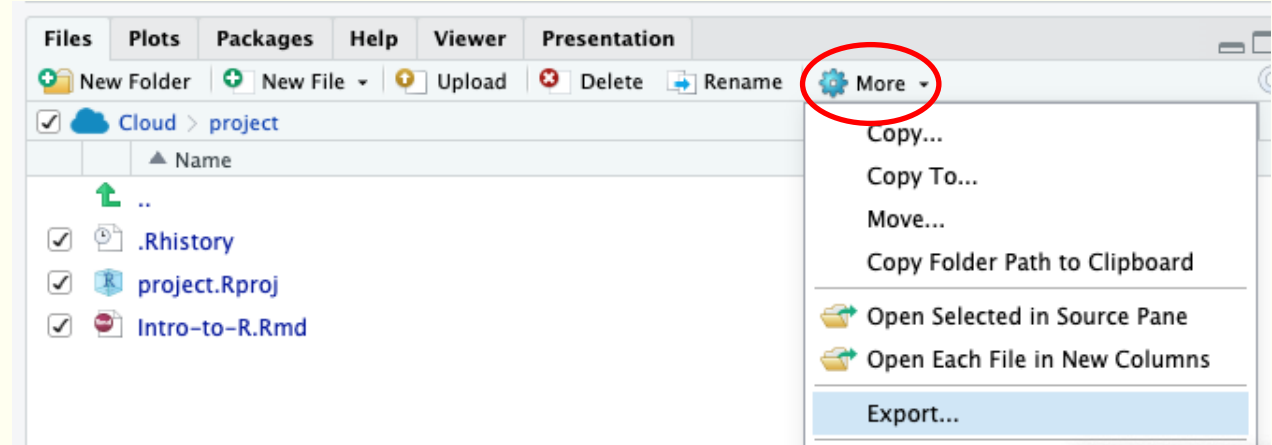
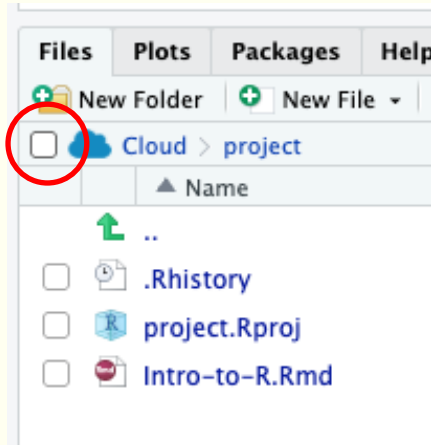


Exporting files out of the CWIC

13

Click on the checkbox on the left of the cloud icon on your *Files* pane

Select the gear icon / "More" in your *Files* pane, and select "Export" in the dropdown



You can unzip this file on your computer and click the .Rproj to open R with all the associated files automatically

Stopping Rstudio

14

- Go back to the Galaxy instance
- Click on the stop button on the Rstudio interactive tool

The screenshot displays the Galaxy CFDE Cloud interface. On the left is a sidebar with navigation options: Upload, Tools, Workflows, Workflow Invocations, Interactive Tools (selected), Visualization, Histories, and History Multiview. The main area is titled 'Interactive Tools' and shows two tools: 'Interactive JupyterLab Notebook' and 'RStudio R 4.4.2 with Bioconductor 3.29'. The 'RStudio' tool is selected, showing its parameters: 'Include data into the environment' (optional) with a 'No datasets available' message, and 'Additional Options' with 'Email notification' and 'Attempt to re-use jobs with identical parameters?' both set to 'No'. There is a 'Run Tool' button. On the right, a 'History' panel shows a list of jobs. Job 11, 'RStudio', is highlighted with a red box around its stop button (a square icon with a red 'X'). Below it, job 16, 'RStudio', is also visible. The top of the interface shows 'Using 11% of 1.0 TB' and the user 'dilliana'.



Q&A